

The scientific level of evidence underlying clinical guideline recommendations across medical specialties

Protocol for a systematic review

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Kim Boesen,¹ Christian Gluud ^{1, 2}

¹ Copenhagen Trial Unit, Centre for Clinical Intervention Research, The Capital Region, Copenhagen University Hospital – Rigshospitalet, Copenhagen, Denmark.

² Department of Regional Health Research, The Faculty of Health Sciences, University of Southern Denmark, Odense, Denmark

Corresponding author

Kim Boesen, MD PhD

kim.boesen@ctu.dk

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Introduction

Over the past 15 years, multiple studies have assessed the scientific evidence underlying guideline recommendations across specialties, countries, and settings (Table 1). Most of these assessments have operated with a 3-tier (or similar) level of evidence hierarchy: level A/I (randomised clinical trials or systematic reviews of such trials); level B/II (single randomised clinical trials or observational research), and level C/III (case studies, expert opinion, standard care). In most of these studies, around half of the recommendations were based on the lowest level of evidence, usually expert opinion or standard care, and only about 15% were based on the highest level of evidence, i.e. multiple randomised clinical trials or systematic reviews of multiple randomised clinical trials (Table 1).

We want to establish an exhaustive list of all (published) assessments of the underlying scientific evidence in clinical guidelines, the Open Guidelines' Assessment Project, through a systematic search and assessment. As of May 2024, we have knowledge of 32 studies, all pointing in the same direction regarding the overall proportion of recommendations supported by highest level evidence (median of 15%), and the proportion of the strongest recommendations supported by randomised clinical trials (median of 21%) (Table 1).

Methods

Database search and screening strategy

We will search PubMed using an exhaustive list of key words based on the known published studies. We will use the artificial intelligence (AI) assisted screening tool ASReview (<https://asreview.nl/>) to identify eligible studies. We will train the ASReview machine learning algorithm on the sample of known studies. One reviewer will screen the obtained references and we will apply an arbitrary stopping rule of 100 consecutive irrelevant records, i.e. we may not screen the full list of obtained records. We will also search the reference lists of all included studies.

Inclusion criteria

1. Any published assessments of clinical guidelines, which have assessed the underlying scientific evidence supporting the individual recommendations, usually referred to as “level of evidence”.
2. Regardless of clinical specialty, subspecialty, or disease, or used methodology.
3. Regardless of year of publication or language.

Outcomes

1. Reported levels of evidence (number of guidelines, number of recommendations, and stratified into evidence levels).
2. Proportion of the strongest recommendations supported with evidence of the highest level (i.e. level A/I).
3. Changes over time in proportion of evidence levels, if assessed at different time points.

Evidence synthesis

We will use descriptive statistics and report overall proportions of the evidence levels with corresponding ranges. To do so, we will (try to) standardise the studies’ evidence definitions, whether they used an A-B-C or an I-II-III system to synthesise the study results. Some studies used other hierarchies, e.g. a 5-tier hierarchy (A-B-C-D-ungraded) or a subdivided 3-tier system, e.g. Class I, IIa, IIb, IIc, III. We will collapse these as appropriate into a 3-tier system and rapport accordingly.

We may encounter a unit-of-analysis issue, since multiple studies assessed overlapping guidelines, hence a global median proportion may be misleading. Similarly, a crude non-weighted study-level mean may also be misleading, as some assessments have included several thousand recommendations and others just a few hundreds.

Discussion

To our knowledge, the underlying scientific evidence in clinical guidelines has not been assessed in a systematic review to estimate a global median proportion of recommendations

supported by high level evidence, i.e. multiple randomised clinical trials or systematic reviews of such trials. It will be important to make such study as it will be a valuable metric to monitor the global development of evidence-based medicine implemented in clinical guidelines.

Data sharing statement

All underlying data will be shared at the Open Guideline Assessment Project website on Zenodo (<https://doi.org/10.5281/zenodo.11144588>).

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Conflicts of interest

None known.

Contributions

KB drafted the first version, CG revised the manuscript, and both authors agreed on the final version to be posted on Zenodo.

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Table 1. Studies assessing levels of evidence backing recommendations in guidelines (chronological order).

Study	Field	Guidelines	Years covering	Evidence level definitions	Evidence levels ('highest' – 'moderate' – 'lowest')	% strongest recommendations backed by 'highest level' evidence
Tricoci (2009)	Cardiology	American College of Cardiology/ American Heart Association	1984 to 2008	A ("Recommendation based on evidence from multiple randomized trials or meta-analyses"). B ("Recommendation based on evidence from a single randomized trial or nonrandomized studies"). C ("Recommendation based on expert opinion, case studies, or standards of care").	11% - 41% - 48%	19%
Lee (2011)	Infectious diseases	Infectious Diseases Society of America	1994 to 2010	I ("≥1 properly randomized controlled trial"). II ("≥1 well-designed clinical trial, without randomization, from cohort or case-controlled analytical studies or from dramatic results from uncontrolled experiments"). III ("opinions of respected authorities based on clinical experience, descriptive studies, or reports of expert committees").	14% - 31% - 55%	23%

Poonancha (2011)	Oncology	National Comprehensive Cancer Network	1996 to 2010	I ("High level of evidence such as randomized controlled trials with uniform consensus"). IIA ("Lower level of evidence with uniform consensus"). IIB ("Lower level of evidence without uniform consensus but no major disagreement"). III ("Any level of evidence but with major disagreement").	6% - 94% - 1% (IIA and IIB collapsed)	Strength of recommendations not reported
Sharma (2011)	Oncology – supportive care	NCCN Supportive care	Not stated to 2011	Same as Poonancha (2011)	5% - 95% - 0% (IIA and IIB collapsed)	Strength of recommendations not reported
Wright (2011)	Obstetrics and gynaecology	American College of Obstetricians and Gynecologists	1999 to 2011	A ("Good and consistent evidence"). B ("Limited or inconsistent evidence"). C ("Consensus and opinion").	30% - 32% - 38%	Strength of recommendations not reported
Cruz (2012)	Oncology – palliative care	NCCN Palliative care	Not stated to 2011	Same as Poonancha (2011)	3% - 97% - 0% (IIA and IIB collapsed)	Strength of recommendations not reported
Rowe (2012)	Hepatology	American Association for the Study of Liver Diseases, European Association for the Study of the Liver, Asian Pacific Association for	2000 to 2011	High quality ("Largely defined as that derived from multiple randomized controlled trials"). Moderate quality ("Evidence as that reported from single randomized controlled trials or nonrandomized studies"). Low quality ("Generated from case	22% - 34% - 44%	Strength of recommendations not reported

		the Study of the Liver		reports and consensus opinions of experts”).		
Koh (2013)	Hepatology	American Association for the Study of Liver Diseases	1998 to 2012	I (“Data derived from multiple randomized controlled trials, or meta-analysis, involving a number of participants to be of statistical power and where further research is unlikely to change the confidence in the estimate of clinical effect”). II (“Data derived from a single randomized trial or nonrandomized studies, cohort or case control analytic studies, and multiple time series where further research may change confidence in the estimate of the clinical effect”). III (“Evidence based on clinical experience, descriptive studies, opinion of respected authorities where further research is very likely to impact confidence on the estimate of clinical effect”).	16% - 44% - 40%	Strength of recommendations not reported
Feuerstein (2013)	Inflammatory bowel disease	Six US, UK, and Canadian associations	1997 to 2011	Grade A (“Randomized controlled trials/meta-analysis”). Grade B (“Single randomized control / nonrandomized”). Grade C (“Expert opinion/case studies/standard of care”).	23% - 25% - 36% - 13%	Strength of recommendations not reported

				Grade D (“No grade of evidence provided for the recommendation”).		
Prusova 2014	Gynaecology and obstetrics	Royal College of Obstetricians and Gynaecologists (UK)	2003 to 2011	<u>Before 2007</u> A (“Literature of overall good quality and evidence”). B (“Well controlled clinical studies”). C (“Evidence from expert committee reports or opinions”). ✓ (“Recommended best practice based on the clinical experience of the guideline development group”). <u>After 2007</u> A (“Meta-analysis, systematic review or a good quality randomised controlled trial”). B (“High quality systematic reviews of case–control or cohort studies”). C (“High quality case–control or cohort studies with a low risk of confounding”). D (“Non- analytical studies, such as case reports/expert opinion”). ✓ (“Recommended best practice based on the clinical experience of the guideline group”).	12% – 22% - 29% - 37% (before 2007) 9% - 12% - 17% - 22% - 40% (after 2007)	Strength of recommendations not reported

Feuerstein (2014)	Interventional medicine ^c	Four US subspecialty associations	Not stated to 2012	Same method as Feuerstein (2013).	11% - 42% - 48%	Strength of recommendations not reported
Baird (2014)	Multiple specialties	Scottish Intercollegiate Guideline Network	Not stated to 2013	1++ ("High quality meta-analyses, systematic reviews of RCTs, or RCTs with a very low risk of bias"). 1+ ("Well-conducted meta-analyses, systematic reviews, or RCTs with a low risk of bias"). 1- ("Meta-analyses, systematic reviews, or RCTs with a high risk of bias"). 2++ ("High quality systematic reviews of case-control or cohort studies"). 2+ ("High quality case-control or cohort studies with a very low risk of confounding or bias and a high probability that the relationship is causal"). 2- ("Case-control or cohort studies with a high risk of confounding or bias and a significant risk that the relationship is not causal"). 3 ("Non-analytic studies eg, case reports and case series"). 4 ("Expert opinion").	24% - 25% - 16% - 36% (sublevels collated)	Strength of recommendations not reported

Alseieri (2015)	Nephrology	Kidney Disease Improving Global Outcomes	2008 to 2013	A ("High level of evidence"). B ("Moderate level of evidence"). C ("Low level of evidence"). D ("Very low level of evidence"). Ungraded Not described in Suppl. 1.	5% - 17% - 31% - 18% - 20%	19%
Ghui (2016)	Gynaecology and obstetrics	The Society of Obstetricians and Gynaecologists of Canada	2000 to 2013	I ("Evidence obtained from at least one properly randomised controlled trial"). II-1 ("Evidence from well-designed controlled trials without randomisation"). II-2 ("Evidence from well-designed cohort (prospective or retrospective) or case-control studies, preferably from more than one centre or research group"). II-3 ("Evidence obtained from comparisons between times or places with or without the intervention. Dramatic results in uncontrolled experiments (such as the results of treatment with penicillin in the 1940s) could also be included in the category"). III ("Opinions of respected authorities, based on clinical experience, descriptive studies, or reports of expert committees").	25% - est. 8% - est. 25% - est. 7% - est. 35% (Exact values reported only for the highest level, other data only reported in pie chart).	57%
Feuerstein (2016)	Osteoarthritis	Various guidelines	Not stated to 2014	Same method as Feuerstein (2013).	35% - 15% - 50%	Strength of recommendations not reported

Feuerstein (2016)	Barrett's esophagitis	Various guidelines	2005 to 2013	Same method as Feuerstein (2013).	6% - 49% - 45%	Strength of recommendations not reported
Jiang 2016	Cough	Various guidelines	2004 to 2016	A ("Randomized controlled trial without important limitations, or meta-analysis"). B (Randomized controlled trial with important limitations, or upgraded observational studies"). C ("Non randomized studies, cohort or case-control studies, case series"). D ("Expert opinion").	12% - 16% - 30% - 41%	Strength of recommendations assessed, but not reported according to level of evidence.
Venkatesh, (2017)	Emergency medicine	American College of Emergency Physicians	1990 to 2016	Class I ("for therapeutic interventions constitutes randomized controlled trials or meta-analysis of randomized controlled trials, while Class I equivalent evidence for diagnostic interventions or prognostication constitutes prospective cohort studies or meta-analyses of the same"). Class II ("nonrandomized trials for therapeutic interventions, or retrospective studies for diagnostic or prognostic actions"). Class III ("case series, case reports, expert consensus, or other reviews").	8% - 32% - 60% (These numbers refer to the cited references in the "current guidelines" (Table 1), and not the recommendations themselves)	Strength of recommendations assessed, but not reported according to level of evidence.
Ebell (2017)	Primary care	Essential Evidence Plus	Not stated to 2017	A ("Consistent and good quality patient-oriented evidence").	18% - 34% - 49%	Strength of recommendations not reported

				B (“Inconsistent or limited quality patient-oriented evidence”). C (“Consensus, usual practice, opinion, disease-oriented evidence or case series”).		
Meyer (2018)	Gastroenterology	American College of Gastroenterology	2007 to 2017	High (“Randomized control trials without limitations, overwhelming evidence from observational studies, systematic reviews, meta-analyses, or if CPG authors concluded that further research was very unlikely to change their confidence in the estimate of effect”). Moderate (“One well-designed clinical trial, randomized control trials with limitations, cohort studies, case-control studies, or if authors believed that further research was likely to have an impact on the confidence in the estimate of effect”). Low (“Expert opinion, clinical experience, descriptive studies, case studies, poor-quality cohort studies, or if the authors expected further research to change the estimate of effect”).	15% - 33% - 52%	Strength of recommendations assessed, but not reported according to level of evidence.
Duarte-Garcia (2018)	Rheumatology	American College of Rheumatology	2010 to 2015	Same method as Tricoci (2009)	23% - 18% - 58%	49%

						(Calculated as 17 of 35 Class I recommendations)
Faranoff (2019)	Cardiology	American College of Cardiology (ACC)/ European Society of Cardiology (ESC)	2008 to 2018	Same method as Tricoci (2009)	ACC: 9% - 50% - 41% ESC: 14% - 31% - 55%	ACC: 14% ESC: 22%
van Dijk (2019)	Cardiology	European Society of Cardiology	2003 to 2018	A ("Multiple randomised clinical trials or meta-analyses"). B ("Single randomised clinical trial or large non-randomised studies"). C ("Expert opinion and/or small studies, retrospective studies, registries").	13% - 31% - 55% (from Figure 1)	21% (300 of 1245 (24%) for therapeutic recommendations; 39 of 831 (5%) for diagnostics).
Schumacher (2019)	Thoracic surgery	American Thoracic Society	Not stated to 2017	High, Medium, Low and very-low Definitions taken from the GRADE system but not reported explicitly in the paper.	9% - 28% - 64%	19%
Venus (2020)	Top 10 causes of death in Australia	Australian guidelines across specialties	2010 to 2018	I ("Systematic review of Level II studies"). II ("RCT or prospective cohort study"). III ("Pseudo-RCT, case-control study, retrospective cohort study, comparative study with or without concurrent controls"). IV ("Case series, study of diagnostic yield, cohort study of persons at different stages of disease or cross-sectional study").	18% - 25% - 29% - 9% - 19%	Strength of recommendations assessed, but not reported according to level of evidence.

				Consensus based (this category was not specified in the method's section)		
Caldeira (2020)	Cardiology	European Society of Cardiology	Not stated to 2019	A-B-C level of evidence system, but definitions not reported. Probably like van Dijk (2019).	12% - 32% - 56%	19% (Figure 1c)
Löhrs (2020)	Psychiatry and neurology	German and Scottish guidelines	Not stated to 2019	Unclear definitions.	<u>Psychiatry and psychotherapy</u> 25% - other evidence levels unclearly reported. <u>Neurology</u> 32% - other evidence levels unclearly reported	Unsure whether reported
Brock (2021)	Maternal and foetal medicine	Society for Maternal-Fetal Medicine (US)	2006 to 2020	A ("High quality evidence"). B ("Moderate-quality evidence"). C ("Low-quality evidence"). Best practice (Levels not specified further in the paper)	15% - 43% - 32% - 10%	21% (34 of 165, calculated from Figure 2)
Weissman (2021)	Colorectal cancer screening	Various guidelines	Not stated to 2020	High quality ("Oxford level 1, GRADE high quality, A level"). Moderate quality ("Oxford level 2, GRADE moderate quality, B level"). Low and very low-quality ("Oxford levels 3, 4, and 5, GRADE low and very low quality, C and D level").	15% - 34% - 51%	Strength of recommendations assessed, but not reported according to level of evidence.
Weissman (2022)	Cancer prevention in IBD	Various guidelines	2005 to 2018	Same methods as Weissman (2021)	7% - 24% - 69%	Strength of recommendations assessed, but not reported according to level of evidence.

Weissman (2022)	Extra-colonic malignancy in IBD	Various guidelines	Not stated to 2020	Same methods as Weissman (2021)	0% - 36% - 64% (note, only 11 recommendations in total)	Strength of recommendations assessed, but not reported according to level of evidence.
Weissman (2023)	Inflammatory bowel disease and nutrition	Various guidelines	Not stated to 2021	Same methods as Weissman (2021)	18% - 26% - 56%	Strength of recommendations assessed, but not reported according to level of evidence.